

HARSHIT SINGH

hs58@illinois.edu | (872)333-2700 | San Diego, CA | linkedin.com/in/harshit-singh-hs27/ | github.com/HarshitSingh27

EDUCATION

University of Illinois Urbana-Champaign <i>Master of Computer Science</i> GPA: 3.8/4	Aug 2023 - May 2025 Champaign, IL, USA
SRM Institute of Science and Technology <i>Bachelor of Technology in Computer Science and Engineering</i> GPA: 3.99/4	Jul 2019 - May 2023 Chennai, INDIA

SKILLS

Programming Languages: Python, Java, SQL
ML & Data: PyTorch, TensorFlow, NumPy, Pandas, Scikit-Learn, NLTK, OpenCV
Databases & Big Data: MySQL, Kafka, MongoDB, Spark, Apache
Web & Frameworks: Node.js, Express.js, Angular, React.js, Next.js, Rest API, Streamlit, Flask, Bootstrap
Cloud & Tools: AWS, Azure, GCP, Git, Linux, Jupyter, Power BI, LangChain, MLflow, Tableau, Kubernetes, Docker
Large Language Models: LLM architecture, Gen AI, RAG, Prompt Engineering

EXPERIENCE

OneFish AI <i>Artificial Intelligence Engineer - Founding</i>	San Diego, CA, USA Aug 2025 - Oct 2025
- Developed a custom Python interpreter and AI compiler that automated complex data workflows, eliminating 95% of manual spreadsheet tasks and scaling to 100K+ records with real time accuracy - Engineered Llama 3 integration with custom APIs under strict resource constraints, optimizing inference to sub second speeds and enabling the model to autonomously decide which AI agent to invoke in real time - Implemented LSTM-based demand forecasting that aligned production with sales patterns, cutting daily waste from 12% to 4% while maximizing operational efficiency	
Palm Beach County <i>Software Development Engineer Intern - AI/ML</i>	West Palm Beach, FL, USA May 2024 - Dec 2024
- Engineered a large-scale RAG pipeline by integrating ElasticSearch, Neural Seek, and Granite 13B on IBM Watson, boosting chatbot accuracy to 96% and serving 1.6M+ residents in a distributed-node environment - Enhanced data infrastructure with B-Tree partitioning, concurrent ingestion, and automated Python-based workflows, cutting manual handling by 80% and reducing overhead by 30% - Implemented enterprise-grade MLOps practices including Oracle RMAN backups and zero-downtime updates, reducing outages by 60% and ensuring fault-tolerant, continuous ML deployment	
Tata Consultancy Services <i>Machine Learning Engineer Intern</i>	Remote, INDIA Jun 2022 - Aug 2022
- Benchmarked Random Forest and XGBoost models for trading signal classification, increasing prediction accuracy by 25% - Developed a multi-tiered data pipeline leveraging AWS (EC2, S3, SageMaker) to process 1M+ daily data points, for scalable and cost-efficient compute in a distributed environment - Reduced LSTM inference latency by 15% using bidirectional LSTM with attention mechanisms, employing batch inference strategies to handle high-volume, real-time time series data	
Paperplane <i>Machine Learning Engineer Intern</i>	Bangalore, INDIA Aug 2021 - Nov 2021
- Fine-tuned a CART-based drug recommendation model using Bayesian optimization, raising prediction accuracy by 11% and streamlining feature selection for diverse clinical requirements - Built a graph-based pipeline (Python, NetworkX) using Pytorch mapping symptom-diagnosis-drug relationships, cutting manual data analysis by 20% and integrating CI/CD (GitHub Actions) for reliable and cost-effective deployment - Leveraged parallel processing with Scikit-Learn and JAX's vmap, boosting classification accuracy by 7% on 1M+ patient records	

Projects

Diversity in Computer Science Faculty PyTorch, Javascript, OpenCV, Facer, Selenium—[Project Link]
- Extracted 4000+ faculty images from 48 universities by developing an automated selenium based web-scraper - Engineered a diversity analytics platform, leveraging OpenCV and Facer for facial feature extraction and demographic analysis
Support Dynamics in Academic Rejection Transformer Models, Hugging Face, Flask, Python—[Project Link]
- Fine-tuned BERT embeddings using TensorFlow with mini-batch optimization and AdamW, reducing training time by 32% and improving sentiment analysis by 6.5% - Built a NLP Flask app, analyzing 900K+ words for sentiment trends from six years of r/gradadmissions, serving 10K+ monthly users.
Disease Probability Prediction Probabilistic Graphical Models, PyTorch, XGBoost, SHAP—[Project Link]
- Achieved 92% accuracy on 25K+ patient records using a Bayesian network (Junction Tree) with XGBoost for probabilistic disease prediction. - Integrated SHAP explainability for model transparency, presented findings at ICRAET-2023 on ML-driven personalized diagnostics